

Chapter 2.2 Probability Distributions

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Plan

1 Review of Probability Distributions

- Normal Distribution
- Chi-Square Distribution
- Student's t-Distribution

2 Multivariate Normal Distribution

- Results Involving Multivariate Normal
- Maximum Likelihood Estimation
- Bivariate normal density

Normal (Gaussian) Distribution

- Bell-shaped distribution with tendency for individuals to clump around the group median/mean
- Used to model many biological phenomena
- Many estimators have approximate normal sampling distributions (see Central Limit Theorem)
- Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$ where μ is mean and σ^2 is variance

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0$$

Obtaining Probabilities and Quantiles in R:

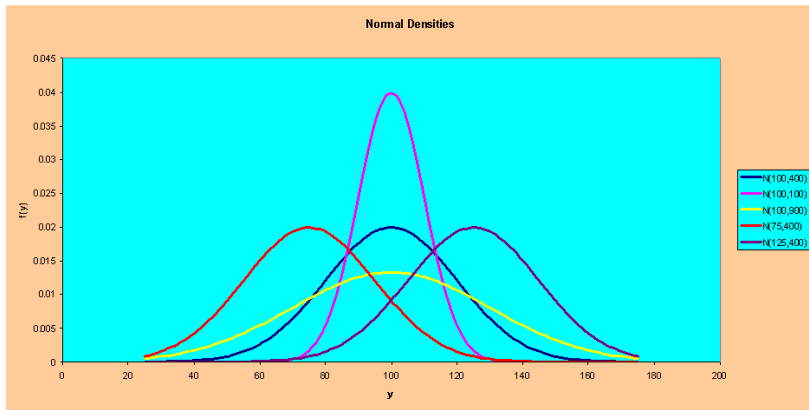
To obtain: $F(x) = P(X \leq x)$; Use Function: `pnorm(x, μ, σ)`

To obtain the p th quantile: $P(X \leq x_p) = p$; Use Function: `qnorm(p, μ, σ)`

Virtually all statistics textbooks give the cdf (or upper tail probabilities) for standardized normal random variables:

$$Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1).$$

Normal Distribution – Density Functions (pdf)



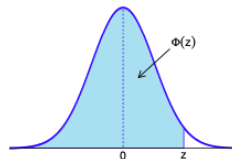
C.1 Phân phối chuẩn tắc (Gauss)

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

Hàng 1: phần lẻ chữ số thứ hai sau dấu thập phân của z .

Cột 1: phần nguyên và một chữ số sau dấu thập phân của z .

Nội dung bảng: giá trị $\Phi(z)$.



Bảng C.1: Giá trị $\Phi(z)$

z	0,00	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,0	0,5000	0,5040	0,5080	0,5120	0,5160	0,5199	0,5239	0,5279	0,5319	0,5359
0,1	0,5398	0,5438	0,5478	0,5517	0,5557	0,5596	0,5636	0,5675	0,5714	0,5753
0,2	0,5793	0,5832	0,5871	0,5910	0,5948	0,5987	0,6026	0,6064	0,6103	0,6141
0,3	0,6179	0,6217	0,6255	0,6293	0,6331	0,6368	0,6406	0,6443	0,6480	0,6517
0,4	0,6554	0,6591	0,6628	0,6664	0,6700	0,6736	0,6772	0,6808	0,6844	0,6879
0,5	0,6915	0,6950	0,6985	0,7019	0,7054	0,7088	0,7123	0,7157	0,7190	0,7224
0,6	0,7257	0,7291	0,7324	0,7357	0,7389	0,7422	0,7454	0,7486	0,7517	0,7549
0,7	0,7580	0,7611	0,7642	0,7673	0,7704	0,7734	0,7764	0,7794	0,7823	0,7852
0,8	0,7881	0,7910	0,7939	0,7967	0,7995	0,8023	0,8051	0,8078	0,8106	0,8133
0,9	0,8159	0,8186	0,8212	0,8238	0,8264	0,8289	0,8315	0,8340	0,8365	0,8389
1,0	0,8413	0,8438	0,8461	0,8485	0,8508	0,8531	0,8554	0,8577	0,8599	0,8621
1,1	0,8643	0,8665	0,8686	0,8708	0,8729	0,8749	0,8770	0,8790	0,8810	0,8830
1,2	0,8849	0,8869	0,8888	0,8907	0,8925	0,8944	0,8962	0,8980	0,8997	0,9015
1,3	0,9032	0,9049	0,9066	0,9082	0,9099	0,9115	0,9131	0,9147	0,9162	0,9177
1,4	0,9192	0,9207	0,9222	0,9236	0,9251	0,9265	0,9279	0,9292	0,9306	0,9319

Chi-Square Distribution

- Indexed by “degrees of freedom (ν)” $X \sim \chi^2_\nu$
- $Z \sim N(0, 1) \Rightarrow Z^2 \sim \chi^2_1$
- Assuming Independence:

$$X_1, \dots, X_n \sim \chi^2_{\nu_i} \quad i = 1, \dots, n \quad \Rightarrow \quad \sum_{i=1}^n X_i \sim \chi^2_{\sum \nu_i}$$

Density Function:

$$f(x) = \frac{1}{\Gamma\left(\frac{\nu}{2}\right) 2^{\nu/2}} x^{(\nu/2)-1} e^{-x/2} \quad x > 0, \nu > 0.$$

$$E\{X\} = \nu \quad \text{and} \quad V\{X\} = 2\nu.$$

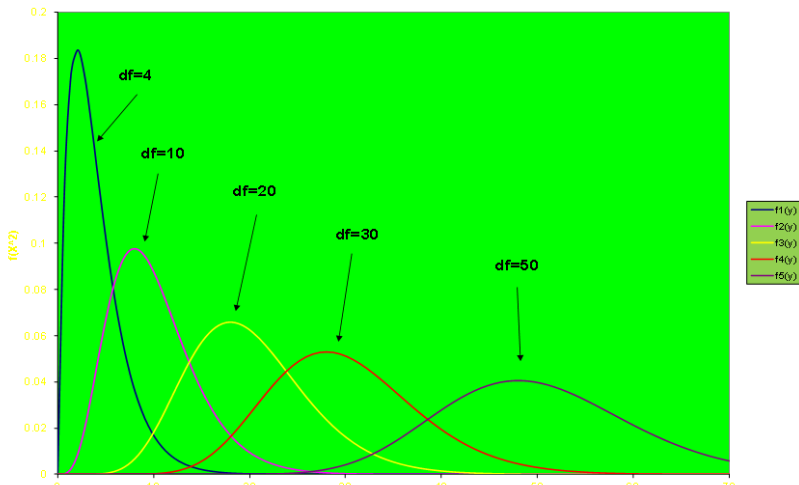
Obtaining Probabilities in R:

To obtain: $1 - F(x) = P(X \geq x)$; Use Function: `pchisq(x,n)`

To obtain quantiles: $P(X \leq x_p) = p$; Use Function: `qchisq(x,n)`

Chi-Square Distributions

Chi-Square Distributions



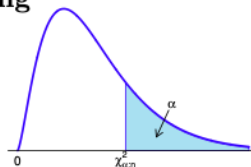
C.3 Phân phối Chi bình phương

$$P(\chi^2 \geq \chi^2_{\alpha;n}) = \alpha \text{ với } \chi^2 \sim \chi^2(n)$$

Hàng 1: giá trị α .

Cột 1: giá trị n .

Nội dung bảng: giá trị $\chi^2_{\alpha;n}$ tương ứng với n và α .



Bảng C.3: Phân phối Chi bình phương

	0,005	0,01	0,015	0,02	0,025	0,03	0,05	0,95	0,975	0,98	0,99	0,995
1	7,879	6,635	5,916	5,412	5,024	4,709	3,841	0,004	0,001	0,001	0,000	0,000
2	10,597	9,210	8,399	7,824	7,378	7,013	5,991	0,103	0,051	0,040	0,020	0,010
3	12,838	11,345	10,465	9,837	9,348	8,947	7,815	0,352	0,216	0,185	0,115	0,072
4	14,860	13,277	12,339	11,668	11,143	10,712	9,488	0,711	0,484	0,429	0,297	0,207
5	16,750	15,086	14,098	13,388	12,833	12,375	11,070	1,145	0,831	0,752	0,554	0,412
6	18,548	16,812	15,777	15,033	14,449	13,968	12,592	1,635	1,237	1,134	0,872	0,676
7	20,278	18,475	17,398	16,622	16,013	15,509	14,067	2,167	1,690	1,564	1,239	0,989
8	21,955	20,090	18,974	18,168	17,535	17,010	15,507	2,733	2,180	2,032	1,646	1,344
9	23,589	21,666	20,513	19,679	19,023	18,480	16,919	3,325	2,700	2,532	2,088	1,735
10	25,188	23,209	22,021	21,161	20,483	19,922	18,307	3,940	3,247	3,059	2,558	2,156
11	26,757	24,725	23,503	22,618	21,920	21,342	19,675	4,575	3,816	3,609	3,053	2,603
12	28,300	26,217	24,963	24,054	23,337	22,742	21,026	5,226	4,404	4,178	3,571	3,074
13	29,819	27,688	26,403	25,472	24,736	24,125	22,362	5,892	5,009	4,765	4,107	3,565
14	31,319	29,141	27,827	26,873	26,119	25,493	23,685	6,571	5,629	5,368	4,660	4,075
15	32,801	30,578	29,235	28,259	27,488	26,848	24,996	7,261	6,262	5,985	5,229	4,601
16	34,267	32,000	30,629	29,633	28,845	28,191	26,296	7,962	6,908	6,614	5,812	5,142
17	35,718	33,409	32,011	30,995	30,191	29,523	27,587	8,672	7,564	7,255	6,408	5,697
18	37,156	34,805	33,382	32,346	31,526	30,845	28,869	9,390	8,231	7,906	7,015	6,265

Student's t-Distribution

- Indexed by “degrees of freedom (n)” $X \sim t_\nu$
- $Z \sim N(0, 1), X \sim \chi_\nu^2$
- Assuming Independence of Z and X :

$$T = \frac{Z}{\sqrt{X/\nu}} \sim t_\nu$$

Density Function:

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad -\infty < t < \infty \quad \nu > 0.$$

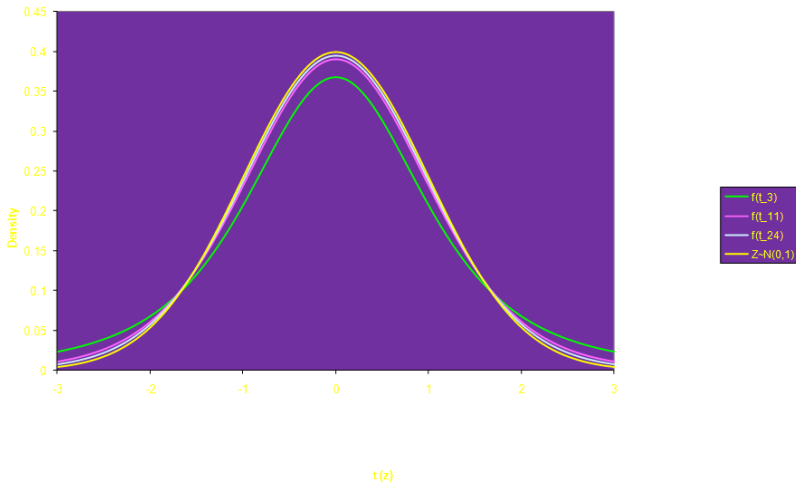
$$E\{T\} = 0 \quad (\nu > 1) \quad \text{and} \quad V\{T\} = \frac{\nu}{\nu-2} \quad (\nu > 2).$$

Obtaining Probabilities /Quantiles in R:

To obtain: $F(t) = P(T \leq t)$: `pt(t,n)`

To obtain: p th quantile: `qt(p,n)`

$t(3)$, $t(11)$, $t(24)$, Z Distributions



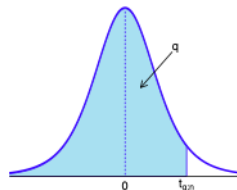
C.2 Phân phối Student

$P(T \leq t_{q;n}) = q$ với $T \sim t(n)$

Hàng 1: giá trị q .

Cột 1: giá trị n .

Nội dung bảng: giá trị $t_{q;n}$ tương ứng với n và q .



Bảng C.2: Phân phối Student

$q \backslash n$	0,60	0,75	0,90	0,95	0,975	0,99	0,995	0,9995
1	0,3249	1,0000	3,0777	6,3138	12,7062	31,8205	63,6567	636,6192
2	0,2887	0,8165	1,8856	2,9200	4,3027	6,9646	9,9248	31,5991
3	0,2767	0,7649	1,6377	2,3534	3,1824	4,5407	5,8409	12,9240
4	0,2707	0,7407	1,5332	2,1318	2,7764	3,7469	4,6041	8,6103
5	0,2672	0,7267	1,4759	2,0150	2,5706	3,3649	4,0321	6,8688
6	0,2648	0,7176	1,4398	1,9432	2,4469	3,1427	3,7074	5,9588
7	0,2632	0,7111	1,4149	1,8946	2,3646	2,9980	3,4995	5,4079
8	0,2619	0,7064	1,3968	1,8595	2,3060	2,8965	3,3554	5,0413
9	0,2610	0,7027	1,3830	1,8331	2,2622	2,8214	3,2498	4,7809
10	0,2602	0,6998	1,3722	1,8125	2,2281	2,7638	3,1693	4,5869
11	0,2596	0,6974	1,3634	1,7959	2,2010	2,7181	3,1058	4,4370
12	0,2590	0,6955	1,3562	1,7823	2,1788	2,6810	3,0545	4,3178

F-Distribution

- Indexed by 2 "degrees of freedom (ν_1, ν_2) " $W \sim F_{\nu_1, \nu_2}$
- $X_1 \sim \chi_{\nu_1}^2, X_2 \sim \chi_{\nu_2}^2$
- Assuming Independence of X_1 and X_2 : $W = \frac{X_1/\nu_1}{X_2/\nu_2} \sim F_{\nu_1, \nu_2}$.

Density Function:

$$f(w) = \frac{\Gamma\left(\frac{\nu_1 + \nu_2}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)} \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} w^{\frac{\nu_1}{2} - 1} \left(\frac{1}{\frac{\nu_1 + \nu_2}{2}}\right) \left(\frac{1}{2}\right)^{-\frac{\nu_1 + \nu_2}{2}} \left[1 + \frac{w\nu_1}{\nu_2}\right]^{-\frac{\nu_1 + \nu_2}{2}}$$

with $w > 0 \quad \nu_1, \nu_2 > 0$.

$$E\{W\} = \frac{\nu_2}{\nu_2 - 2} \quad (\nu_2 > 2)$$

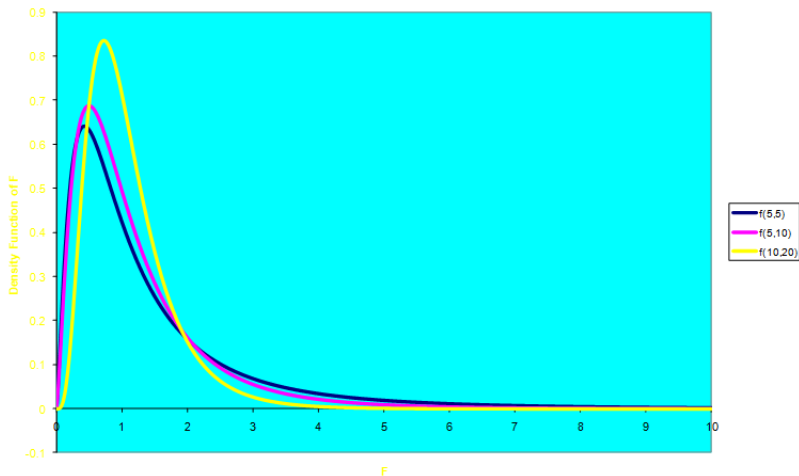
$$V\{W\} = \left(\frac{\nu_2}{\nu_2 - 2}\right)^2 \frac{\nu_1 + \nu_2 - 2}{\nu_1(\nu_2 - 4)} \quad (\nu_2 > 4).$$

Obtaining Probabilities/Quantiles in R:

To obtain: $F(w) = P(W \leq w)$: *pf*(w, ν_1, ν_2)

To obtain: p th quantile: *qf*(p, ν_1, ν_2)

F-Distributions



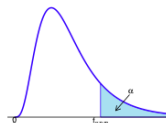
C.4 Phân phối Fisher

$P(F \geq f_{\alpha;n,m}) = \alpha$ với $F \sim F(n, m)$

Hàng 1: giá trị n .

Cột 1: giá trị m .

Nội dung bảng: giá trị $f_{\alpha;n,m}$ tương ứng với n, m và α .



Bảng C.4: $\alpha = 0,05$

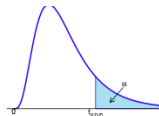
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	$+\infty$
1	161,45	199,50	215,71	224,58	230,16	233,99	236,77	238,88	240,54	241,88	243,91	245,95	248,01	249,05	250,10	251,14	252,20	253,25	254,30
2	18,51	19,00	19,16	19,25	19,30	19,33	19,35	19,37	19,38	19,40	19,41	19,43	19,45	19,45	19,46	19,47	19,48	19,49	19,50
3	10,13	9,55	9,28	9,12	9,01	8,94	8,89	8,85	8,81	8,79	8,74	8,70	8,66	8,64	8,62	8,59	8,57	8,55	8,53
4	7,71	6,94	6,59	6,39	6,26	6,16	6,09	6,04	6,00	5,96	5,91	5,86	5,80	5,77	5,75	5,72	5,69	5,66	5,63
5	6,61	5,79	5,41	5,19	5,05	4,95	4,88	4,82	4,77	4,74	4,68	4,62	4,56	4,53	4,50	4,46	4,43	4,40	4,37
6	5,99	5,14	4,76	4,53	4,39	4,28	4,21	4,15	4,10	4,06	4,00	3,94	3,87	3,84	3,81	3,77	3,74	3,70	3,67
7	5,59	4,74	4,35	4,12	3,97	3,87	3,79	3,73	3,68	3,64	3,57	3,51	3,44	3,41	3,38	3,34	3,30	3,27	3,23
8	5,32	4,46	4,07	3,84	3,69	3,58	3,50	3,44	3,39	3,35	3,28	3,22	3,15	3,12	3,08	3,04	3,01	2,97	2,93
9	5,12	4,26	3,86	3,63	3,48	3,37	3,29	3,23	3,18	3,14	3,07	3,01	2,94	2,90	2,86	2,83	2,79	2,75	2,71
10	4,96	4,10	3,71	3,48	3,33	3,22	3,14	3,07	3,02	2,98	2,91	2,85	2,77	2,74	2,70	2,66	2,62	2,58	2,54
11	4,84	3,98	3,59	3,36	3,20	3,09	3,01	2,95	2,90	2,85	2,79	2,72	2,65	2,61	2,57	2,53	2,49	2,45	2,41
12	4,75	3,89	3,49	3,26	3,11	3,00	2,91	2,85	2,80	2,75	2,69	2,62	2,54	2,51	2,47	2,43	2,38	2,34	2,30
13	4,67	3,81	3,41	3,18	3,03	2,92	2,83	2,77	2,71	2,67	2,60	2,53	2,46	2,42	2,38	2,34	2,30	2,25	2,21
14	4,60	3,74	3,34	3,11	2,96	2,85	2,76	2,70	2,65	2,60	2,53	2,46	2,39	2,35	2,31	2,27	2,22	2,18	2,13
15	4,54	3,68	3,29	3,06	2,90	2,79	2,71	2,64	2,59	2,54	2,48	2,40	2,33	2,29	2,25	2,20	2,16	2,11	2,07
16	4,49	3,63	3,24	3,01	2,85	2,74	2,66	2,59	2,54	2,49	2,42	2,35	2,28	2,24	2,19	2,15	2,11	2,06	2,01
17	4,45	3,59	3,20	2,96	2,81	2,70	2,61	2,55	2,49	2,45	2,38	2,31	2,23	2,19	2,15	2,10	2,06	2,01	1,96
18	4,41	3,55	3,16	2,93	2,77	2,66	2,58	2,51	2,46	2,41	2,34	2,27	2,19	2,15	2,11	2,06	2,02	1,97	1,92
19	4,38	3,52	3,13	2,90	2,74	2,63	2,54	2,48	2,42	2,38	2,31	2,23	2,16	2,11	2,07	2,03	1,98	1,93	1,88
20	4,35	3,49	3,10	2,87	2,71	2,60	2,51	2,45	2,39	2,35	2,28	2,20	2,12	2,08	2,04	1,99	1,95	1,90	1,84
21	4,32	3,47	3,07	2,84	2,68	2,57	2,49	2,42	2,37	2,32	2,25	2,18	2,10	2,05	2,01	1,96	1,92	1,87	1,81
22	4,30	3,44	3,05	2,82	2,66	2,55	2,46	2,40	2,34	2,30	2,23	2,15	2,07	2,03	1,98	1,94	1,89	1,84	1,78
23	4,28	3,42	3,03	2,80	2,64	2,53	2,44	2,37	2,32	2,27	2,20	2,13	2,05	2,01	1,96	1,91	1,86	1,81	1,76
24	4,26	3,40	3,01	2,78	2,62	2,51	2,42	2,36	2,30	2,25	2,18	2,11	2,03	1,98	1,94	1,89	1,84	1,79	1,73
25	4,24	3,39	2,99	2,76	2,60	2,49	2,40	2,34	2,28	2,24	2,16	2,09	2,01	1,96	1,92	1,87	1,82	1,77	1,71
30	4,17	3,32	2,92	2,69	2,53	2,42	2,33	2,27	2,21	2,16	2,09	2,01	1,93	1,89	1,84	1,79	1,74	1,68	1,62
40	4,08	3,23	2,84	2,61	2,45	2,34	2,25	2,18	2,12	2,08	2,00	1,92	1,84	1,79	1,74	1,69	1,64	1,58	1,51
60	4,00	3,15	2,76	2,53	2,37	2,25	2,17	2,10	2,04	1,99	1,92	1,84	1,75	1,70	1,65	1,59	1,53	1,47	1,39
120	3,92	3,07	2,68	2,45	2,29	2,18	2,09	2,02	1,96	1,91	1,83	1,75	1,66	1,61	1,55	1,50	1,43	1,35	1,26
$+\infty$	3,84	3,00	2,61	2,37	2,21	2,10	2,01	1,94	1,88	1,83	1,75	1,67	1,57	1,52	1,46	1,40	1,32	1,22	1,03

$P(F \geq f_{\alpha;n,m}) = \alpha$ với $F \sim F(n, m)$

Hàng 1: giá trị n .

Cột 1: giá trị m .

Nội dung bảng: giá trị $f_{\alpha;n,m}$ tương ứng với n, m và α .



Bảng C.5: $\alpha = 0,01$

	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	+∞
1	4052,184999	505403,355624	585763,655858	59,99	5928,365981	0,76022	476055,856106	326157,286208	73	6234,636260	656286,786313	0,36339	396365,55						
2	98,50	99,00	99,17	99,25	99,30	99,33	99,36	99,37	99,39	99,40	99,42	99,43	99,45	99,46	99,47	99,47	99,48	99,49	99,50
3	34,12	30,82	29,46	28,71	28,24	27,91	27,67	27,49	27,35	27,23	27,05	26,87	26,69	26,60	26,50	26,41	26,32	26,22	26,13
4	21,20	18,00	16,69	15,98	15,52	15,21	14,98	14,80	14,66	14,55	14,37	14,20	14,02	13,93	13,84	13,75	13,65	13,56	13,46
5	16,26	13,27	12,06	11,39	10,97	10,67	10,46	10,29	10,16	10,05	9,89	9,72	9,55	9,47	9,38	9,29	9,20	9,11	9,02
6	13,75	10,92	9,78	9,15	8,75	8,47	8,26	8,10	7,98	7,87	7,72	7,56	7,40	7,31	7,23	7,14	7,06	6,97	6,88
7	12,25	9,55	8,45	7,85	7,46	7,19	6,99	6,84	6,72	6,62	6,47	6,31	6,16	6,07	5,99	5,91	5,82	5,74	5,65
8	11,26	8,65	7,59	7,01	6,63	6,37	6,18	6,03	5,91	5,81	5,67	5,52	5,36	5,28	5,20	5,12	5,03	4,95	4,86
9	10,56	8,02	6,99	6,42	6,06	5,80	5,61	5,47	5,35	5,26	5,11	4,96	4,81	4,73	4,65	4,57	4,48	4,40	4,31
10	10,04	7,56	6,55	5,99	5,64	5,39	5,20	5,06	4,94	4,85	4,71	4,56	4,41	4,33	4,25	4,17	4,08	4,00	3,91
11	9,65	7,21	6,22	5,67	5,32	5,07	4,89	4,74	4,63	4,54	4,40	4,25	4,10	4,02	3,94	3,86	3,78	3,69	3,60
12	9,33	6,93	5,95	5,41	5,06	4,82	4,64	4,50	4,39	4,30	4,16	4,01	3,86	3,78	3,70	3,62	3,54	3,45	3,36
13	9,07	6,70	5,74	5,21	4,86	4,62	4,44	4,30	4,19	4,10	3,96	3,82	3,66	3,59	3,51	3,43	3,34	3,25	3,17
14	8,86	6,51	5,56	5,04	4,69	4,46	4,28	4,14	4,03	3,94	3,80	3,66	3,51	3,43	3,35	3,27	3,18	3,09	3,01
15	8,68	6,36	5,42	4,89	4,56	4,32	4,14	4,00	3,89	3,80	3,67	3,52	3,37	3,29	3,21	3,13	3,05	2,96	2,87
16	8,53	6,23	5,29	4,77	4,44	4,20	4,03	3,89	3,78	3,69	3,55	3,41	3,26	3,18	3,10	3,02	2,93	2,84	2,75
17	8,40	6,11	5,18	4,67	4,34	4,10	3,93	3,79	3,68	3,59	3,46	3,31	3,16	3,08	3,00	2,92	2,83	2,75	2,65
18	8,29	6,01	5,09	4,58	4,25	4,01	3,84	3,71	3,60	3,51	3,37	3,23	3,08	3,00	2,92	2,84	2,75	2,66	2,57
19	8,18	5,93	5,01	4,50	4,17	3,94	3,77	3,63	3,52	3,43	3,30	3,15	3,00	2,92	2,84	2,76	2,67	2,58	2,49
20	8,10	5,85	4,94	4,43	4,10	3,87	3,70	3,56	3,46	3,37	3,23	3,09	2,94	2,86	2,78	2,69	2,61	2,52	2,42
21	8,02	5,78	4,87	4,37	4,04	3,81	3,64	3,51	3,40	3,31	3,17	3,03	2,88	2,80	2,72	2,64	2,55	2,46	2,36
22	7,95	5,72	4,82	4,31	3,99	3,76	3,59	3,45	3,35	3,26	3,12	2,98	2,83	2,75	2,67	2,58	2,50	2,40	2,31
23	7,88	5,66	4,76	4,26	3,94	3,71	3,54	3,41	3,30	3,21	3,07	2,93	2,78	2,70	2,62	2,54	2,45	2,35	2,26
24	7,82	5,61	4,72	4,22	3,90	3,67	3,50	3,36	3,26	3,17	3,03	2,89	2,74	2,66	2,58	2,49	2,40	2,31	2,21
25	7,77	5,57	4,68	4,18	3,85	3,63	3,46	3,32	3,22	3,13	2,99	2,85	2,70	2,62	2,54	2,45	2,36	2,27	2,17
30	7,56	5,39	4,51	4,02	3,70	3,47	3,30	3,17	3,07	2,98	2,84	2,70	2,55	2,47	2,39	2,30	2,21	2,11	2,01
40	7,31	5,18	4,31	3,83	3,51	3,29	3,12	2,99	2,89	2,80	2,66	2,52	2,37	2,29	2,20	2,11	2,02	1,92	1,81
60	7,08	4,98	4,13	3,65	3,34	3,12	2,95	2,82	2,72	2,63	2,50	2,35	2,20	2,12	2,03	1,94	1,84	1,73	1,60
120	6,85	4,79	3,95	3,48	3,17	2,96	2,79	2,66	2,56	2,47	2,34	2,19	2,03	1,95	1,86	1,76	1,66	1,53	1,38
+∞	6,64	4,61	3,78	3,32	3,02	2,80	2,64	2,51	2,41	2,32	2,19	2,04	1,88	1,79	1,70	1,59	1,48	1,33	1,05

Plan

1 Review of Probability Distributions

- Normal Distribution
- Chi-Square Distribution
- Student's t-Distribution

2 Multivariate Normal Distribution

- Results Involving Multivariate Normal
- Maximum Likelihood Estimation
- Bivariate normal density

Multivariate Normal Distribution

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix} \quad \mu_{\mathbf{X}} = \mathbf{E}\{\mathbf{X}\} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix} \quad \Sigma_{\mathbf{X}} = \Sigma^2\{\mathbf{X}\} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \sigma_{21} & \cdots & \sigma_{2p} \\ \vdots & \ddots & \vdots \\ \sigma_{1p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Multivariate Normal Density function: $\mathbf{X} \sim \mathcal{N}_p(\mu_{\mathbf{X}}, \Sigma_{\mathbf{X}})$

$$f(\mathbf{X}) = (2\pi)^{-p/2} |\Sigma_{\mathbf{X}}|^{-1/2} \exp \left[-\frac{1}{2} (\mathbf{x} - \mu_{\mathbf{X}})' \Sigma_{\mathbf{X}}^{-1} (\mathbf{x} - \mu_{\mathbf{X}}) \right]$$

$$\Rightarrow X_i \sim \mathcal{N}(\mu_i, \sigma_{ii}) \quad i = 1, \dots, p \quad \sigma\{X_i, X_j\} \equiv \sigma_{ij} \quad i \neq j.$$

Note, if $\mathbf{A}$ is a (full rank) matrix of fixed constants:

$$\mathbf{W} = \mathbf{AX} \sim \mathcal{N}(\mathbf{A}\mu_{\mathbf{X}}, \mathbf{A}\Sigma_{\mathbf{X}}\mathbf{A}')$$

Results Involving Multivariate Normal - I

Result 1

If $\Sigma_{\mathbf{X}}$ is positive definite with eigenvalue/eigenvector pairs $(\lambda_i, \mathbf{e}_i)$
 Then $\Sigma_{\mathbf{X}}^{-1}$ is positive definite with eigenvalue/eigenvector pairs $(1/\lambda_i, \mathbf{e}_i)$
 $i = 1, \dots, p$.

$$\Rightarrow \Sigma_{\mathbf{X}} = \sum_{i=1}^n \lambda_i \mathbf{e}_i \mathbf{e}_i' = \mathbf{P} \mathbf{\Lambda} \mathbf{P}' \quad \Sigma_{\mathbf{X}}^{-1} = \sum_{i=1}^n \left(\frac{1}{\lambda_i} \right) \mathbf{e}_i \mathbf{e}_i' = \mathbf{P} \mathbf{\Lambda}^{-1} \mathbf{P}'$$

$$\text{where } \mathbf{P} = \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_p \end{bmatrix} \quad ; \quad \mathbf{\Lambda} = \text{diag} \{ \lambda_i \}.$$

Contours of constant density for p -dim MVN are ellipsoids wrt $\mathbf{X}$ s.t.

$$(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})' \Sigma_{\mathbf{X}}^{-1} (\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}) = c^2$$

Ellipsoids centered at $\boldsymbol{\mu}_{\mathbf{X}}$ with axes $\pm c \sqrt{\lambda_i} \mathbf{e}_i$, $i = 1, \dots, p$.

Setting $c^2 = \chi_p^2(\alpha)$ s.t. $P(\chi_p^2 \geq \chi_p^2(\alpha)) = \alpha$

$\Rightarrow$ the solid ellipsoid $(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})' \Sigma_{\mathbf{X}}^{-1} (\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}) \leq \chi_p^2(\alpha)$ has probability $1 - \alpha$.

$\text{Cov} \{X_i, X_j\} = 0 \Rightarrow X_i, X_j \text{ independent.}$

Results Involving Multivariate Normal - II

Result 2

If $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$, then any linear combination of variable

$$\mathbf{a}'\mathbf{X} = a_1X_1 + a_2X_2 + \cdots + a_pX_p \sim \mathcal{N}(\mathbf{a}\boldsymbol{\mu}_X, \mathbf{a}'\boldsymbol{\Sigma}_X\mathbf{a}).$$

Also, if $\mathbf{a}'\mathbf{X} \sim \mathcal{N}(\mathbf{a}\boldsymbol{\mu}_X, \mathbf{a}'\boldsymbol{\Sigma}_X\mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$.

Results Involving Multivariate Normal - III

Result 3

If $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$ then the q linear combinations

$$\mathbf{AX} = \begin{bmatrix} a_{11}X_1 + \cdots + a_{1p}X_p \\ a_{21}X_1 + \cdots + a_{2p}X_p \\ \vdots \\ a_{q1}X_1 + \cdots + a_{qp}X_p \end{bmatrix} \sim \mathcal{N}_q(\mathbf{A}\boldsymbol{\mu}_X, \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}').$$

Also,

$$\mathbf{X} + \mathbf{d} \sim \mathcal{N}_p(\boldsymbol{\mu}_X + \mathbf{d}, \boldsymbol{\Sigma}_X)$$

where $\mathbf{d}_{p \times 1}$ is a vector of constants.

Result 4

All subsets of $\mathbf{X}$ are normally distributed. If we respectively partition $\mathbf{X}$, its mean vector $\boldsymbol{\mu}$ and its covariance matrix $\boldsymbol{\Sigma}$ as

$$\mathbf{X}_{(p \times 1)} = \left[\begin{array}{c} \mathbf{X}_1 \\ \hline \mathbf{X}_2 \end{array} \right] \quad \mu_{(p \times 1)} = \left[\begin{array}{c} \mu_1 \\ \hline \mu_2 \end{array} \right]$$

$$\Sigma_{(p \times p)} = \left[\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \hline \Sigma_{21} & \Sigma_{22} \end{array} \right]$$

then $\mathbf{X}_1 \sim \mathcal{N}_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$.

Results Involving Multivariate Normal - V

Result 5

(a). If $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent, then $\text{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$.

$(q_1 \times 1)$ $(q_2 \times 1)$ $(q_1 \times q_2)$

(b). If $\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim \mathcal{N}_{q_1+q_2} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$,
then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent if and only if $\Sigma_{12} = \mathbf{0}$.

(c). If $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent and distributed as $\mathcal{N}_{q_1}(\mu_1, \Sigma_{11})$ and $\mathcal{N}_{q_2}(\mu_2, \Sigma_{22})$ respectively, then

$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim \mathcal{N}_{q_1+q_2} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \mathbf{0} \\ \mathbf{0} & \Sigma_{22} \end{bmatrix} \right).$$

Results Involving Multivariate Normal - VI

Result 6: Conditional Distributions of components are MVN:

$$(\mathbf{X}_1 \equiv q \times 1, \quad \mathbf{X}_2 \equiv (p - q) \times 1)$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim \mathcal{N}_p(\boldsymbol{\mu}_\mathbf{X}, \boldsymbol{\Sigma}_\mathbf{X}) \text{ with } \boldsymbol{\mu}_\mathbf{X} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \boldsymbol{\Sigma}_\mathbf{X} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \quad \text{s.t. } |\boldsymbol{\Sigma}_{22}| > 0$$

$$\Rightarrow \mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2 \sim \mathcal{N}_q(\boldsymbol{\mu}_{1|\mathbf{x}_2}; \boldsymbol{\Sigma}_{11|\mathbf{x}_2})$$

with

$$\text{Mean: } \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2) = \boldsymbol{\mu}_{1|\mathbf{x}_2}$$

$$\text{Variance-Covariance Matrix: } \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{11|\mathbf{x}_2}.$$

$$\Rightarrow f(\mathbf{X}_1 | \mathbf{x}_2) = (2\pi)^{-q/2} |\boldsymbol{\Sigma}_{11|\mathbf{x}_2}|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{x}_1 - \boldsymbol{\mu}_{1|\mathbf{x}_2})' \boldsymbol{\Sigma}_{11|\mathbf{x}_2}^{-1} (\mathbf{x}_1 - \boldsymbol{\mu}_{1|\mathbf{x}_2}) \right\}$$

where $x_{1i} \in \mathbb{R}; i = 1, \dots, q$.

Note that the conditional mean depends on the specific level(s) $\mathbf{x}_2$, the conditional variance does not.

Results Involving Multivariate Normal - VI

Special Case: $p = 2, q = 1$

$$X_1|X_2 = x_2 \sim \mathcal{N}(\mu_{1|x_2}, \sigma_{11|x_2})$$

with

$$\mu_{1|x_2} = \mu_1 + \frac{\sigma_{12}}{\sigma_{22}} (x_2 - \mu_2)$$

$$\sigma_{11|x_2} = \sigma_{11} - \frac{\sigma_{12}^2}{\sigma_{22}} = \sigma_{11} - \left(\frac{\sigma_{11}}{\sigma_{11}}\right) \frac{\sigma_{12}^2}{\sigma_{22}} = \sigma_{11} (1 - \rho_{12}^2)$$

$$\Rightarrow f(x_1, x_2) = \frac{1}{\sqrt{2\pi\sigma_{11}(1-\rho_{12}^2)}} \exp\left\{-\frac{1}{2\sigma_{11}(1-\rho_{12}^2)}\left(x_1 - \left(\mu_1 + \frac{\sigma_{12}}{\sigma_{22}}(x_2 - \mu_2)\right)\right)^2\right\}$$

$$-\infty < x_1 < \infty.$$

Results Involving Multivariate Normal - VII

Result 7

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be mutually independent with $\mathbf{X}_j \sim \mathcal{N}_p(\boldsymbol{\mu}_j, \boldsymbol{\Sigma})$. Then

$$\mathbf{W}_1 = a_1 \mathbf{X}_1 + \dots + a_n \mathbf{X}_n \sim \mathcal{N}_p \left(\sum_{j=1}^n a_j \boldsymbol{\mu}_j, \left(\sum_{j=1}^n a_j^2 \right) \boldsymbol{\Sigma} \right)$$

$$\mathbf{W}_2 = b_1 \mathbf{X}_1 + \dots + b_n \mathbf{X}_n \sim \mathcal{N}_p \left(\sum_{j=1}^n b_j \boldsymbol{\mu}_j, \left(\sum_{j=1}^n b_j^2 \right) \boldsymbol{\Sigma} \right)$$

$\mathbf{W}_1, \mathbf{W}_2$ are jointly distributed normal with

$$\boldsymbol{\mu}_{\mathbf{W}_1 \mathbf{W}_2} = \begin{bmatrix} \sum_{j=1}^n a_j \boldsymbol{\mu}_j \\ \sum_{j=1}^n b_j \boldsymbol{\mu}_j \end{bmatrix} \text{ and } \boldsymbol{\Sigma}_{\mathbf{W}_1 \mathbf{W}_2} = \begin{bmatrix} \left(\sum_{j=1}^n a_j^2 \right) \boldsymbol{\Sigma} & \left(\sum_{j=1}^n a_j b_j \right) \boldsymbol{\Sigma} \\ \left(\sum_{j=1}^n a_j b_j \right) \boldsymbol{\Sigma} & \left(\sum_{j=1}^n b_j^2 \right) \boldsymbol{\Sigma} \end{bmatrix}$$

Consequently, $\mathbf{W}_1$ and $\mathbf{W}_2$ are independent if $\sum_{j=1}^n a_j b_j = 0$.

Multivariate Normal Likelihood Function I

$\mathbf{X}_1, \dots, \mathbf{X}_n \equiv$ random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with **joint density**:

$$\prod_{j=1}^n \left[(2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{X}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X}_j - \boldsymbol{\mu}) \right\} \right] = \\ (2\pi)^{-np/2} |\boldsymbol{\Sigma}|^{-n/2} \exp \left\{ -\frac{1}{2} \sum_{j=1}^n (\mathbf{X}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X}_j - \boldsymbol{\mu}) \right\}.$$

Exponential term multiplied by $-1/2$:

$$\sum_{j=1}^n (\mathbf{X}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X}_j - \boldsymbol{\mu}) = \text{tr} \left\{ \sum_{j=1}^n (\mathbf{X}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X}_j - \boldsymbol{\mu}) \right\} = \\ \text{tr} \left\{ \sum_{j=1}^n \boldsymbol{\Sigma}^{-1} (\mathbf{X}_j - \boldsymbol{\mu}) (\mathbf{X}_j - \boldsymbol{\mu})' \right\} = \text{tr} \left\{ \boldsymbol{\Sigma}^{-1} \sum_{j=1}^n (\mathbf{X}_j - \boldsymbol{\mu}) (\mathbf{X}_j - \boldsymbol{\mu})' \right\}.$$

Multivariate Normal Likelihood Function II

$$\begin{aligned}
\sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}) (\mathbf{x}_j - \boldsymbol{\mu})' &= \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}} + \bar{\mathbf{X}} - \boldsymbol{\mu}) (\mathbf{x}_j - \bar{\mathbf{X}} + \bar{\mathbf{X}} - \boldsymbol{\mu})' = \\
&\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}}) (\mathbf{x}_j - \bar{\mathbf{X}})' + \sum_{j=1}^n (\bar{\mathbf{X}} - \boldsymbol{\mu}) (\bar{\mathbf{X}} - \boldsymbol{\mu})' + 2 \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}}) (\bar{\mathbf{X}} - \boldsymbol{\mu})' \\
&= \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}}) (\mathbf{x}_j - \bar{\mathbf{X}})' + \sum_{j=1}^n (\bar{\mathbf{X}} - \boldsymbol{\mu}) (\bar{\mathbf{X}} - \boldsymbol{\mu})' + \\
&2 \left(\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}}) \right) (\bar{\mathbf{X}} - \boldsymbol{\mu})' \\
&= \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}}) (\mathbf{x}_j - \bar{\mathbf{X}})' + n (\bar{\mathbf{X}} - \boldsymbol{\mu}) (\bar{\mathbf{X}} - \boldsymbol{\mu})' + 0.
\end{aligned}$$

Joint Density / Likelihood Function:

$$(2\pi)^{-np/2} |\boldsymbol{\Sigma}|^{-n/2} \exp \left\{ -\frac{1}{2} \text{tr} \left\{ \boldsymbol{\Sigma}^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}}) (\mathbf{x}_j - \bar{\mathbf{X}})' + n (\bar{\mathbf{X}} - \boldsymbol{\mu}) (\bar{\mathbf{X}} - \boldsymbol{\mu})' \right] \right\} \right\}.$$

Maximum Likelihood Estimator of μ

Likelihood Function:

$$\begin{aligned}
 & (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \text{tr} \left\{ \Sigma^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' + n (\bar{\mathbf{x}} - \mu) (\bar{\mathbf{x}} - \mu)' \right] \right\} \right\} \\
 &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \left[\text{tr} \left\{ \Sigma^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' \right] \right\} + \text{tr} \left\{ \Sigma^{-1} n (\bar{\mathbf{x}} - \mu) (\bar{\mathbf{x}} - \mu)' \right\} \right] \right\} \\
 &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \left[\text{tr} \left\{ \Sigma^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' \right] \right\} + n (\bar{\mathbf{x}} - \mu)' \Sigma^{-1} (\bar{\mathbf{x}} - \mu) \right] \right\}
 \end{aligned}$$

Maximum Likelihood Estimator for μ :

$$\begin{aligned}
 L(\mu, \Sigma) &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \left[\text{tr} \left\{ \Sigma^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' \right] \right\} \right] \right\} \exp \left\{ -\frac{1}{2} \left[n (\bar{\mathbf{x}} - \mu)' \Sigma^{-1} (\bar{\mathbf{x}} - \mu) \right] \right\}
 \end{aligned}$$

maximized when $\hat{\mu} = \bar{\mathbf{x}}$

$$\Rightarrow \exp \left\{ -\frac{1}{2} \left[n (\bar{\mathbf{x}} - \mu)' \Sigma^{-1} (\bar{\mathbf{x}} - \mu) \right] \right\} = 1$$

is at its maximum since Σ^{-1} is positive definite.

Maximum Likelihood Estimator of Σ

Result: $\mathbf{B} \equiv p \times p$ positive definite, scalar $b > 0$, $\Sigma \equiv$ positive definite:

$$\frac{1}{|\Sigma|^b} \exp \left\{ -\frac{1}{2} \text{tr} \{ \Sigma^{-1} \mathbf{B} \} \right\} \leq \frac{1}{|\mathbf{B}|^b} (2b)^{bp} e^{-bp} \text{ with equality holding at } \Sigma = \left(\frac{1}{2b} \right) \mathbf{B}$$

Maximum Likelihood Estimator for Σ evaluated at $\hat{\mu}$:

$$L(\mu, \Sigma) = (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \left[\text{tr} \left\{ \Sigma^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' \right] \right\} \right] \right\}$$

$$\text{setting } b = \frac{n}{2} \quad \mathbf{B} = \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})'$$

$$\Rightarrow \hat{\Sigma} = \left(\frac{1}{2(n/2)} \right) \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' = \frac{1}{n} \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' = \frac{n-1}{n} \mathbf{S}$$

Likelihood Function evaluated at the observed ML estimates: $L(\hat{\mu}, \hat{\Sigma}) =$

$$(2\pi)^{-np/2} |\hat{\Sigma}|^{-n/2} \exp \left\{ -\frac{1}{2} \left[\text{tr} \left\{ (\hat{\Sigma})^{-1} \left[\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' \right] \right\} \right] \right\} =$$

$$= (2\pi)^{-np/2} |\hat{\Sigma}|^{-n/2} e^{-np/2}. \text{ Note: } |\hat{\Sigma}| = \left| \frac{n-1}{n} \mathbf{S} \right| = \left(\frac{n-1}{n} \right)^p |\mathbf{S}|$$

$$\Rightarrow L(\hat{\mu}, \hat{\Sigma}) = (2\pi)^{-np/2} e^{-np/2} \left(\frac{n-1}{n} \right)^p |\mathbf{S}| = \text{constant} \times \text{generalized inverse.}$$

Results for ML Estimators and Large - Sample Properties I

$\theta \equiv$ Parameter vector; $h(\theta) \equiv$ function of θ ;

ML Estimate of $h(\theta) \equiv h\left(\hat{\theta}\right)$.

Sufficient Statistics: Joint density of $\mathbf{X}_1, \dots, \mathbf{X}_n$ depends only on the observed data through: $\bar{\mathbf{X}}$ and $(n-1)\mathbf{S} = \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{X}})' (\mathbf{x}_j - \bar{\mathbf{X}})$

$\Rightarrow \bar{\mathbf{X}}$ and $\mathbf{S}$ are Sufficient Statistics.

Sampling Distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$ under Normality and Independence:

- $\bar{\mathbf{X}} \sim \mathcal{N}_p\left(\mu, \frac{1}{n}\Sigma\right)$
- $(n-1)\mathbf{S} \sim$ Wishart with d.f. = $n-1$
- $\bar{\mathbf{X}}$ and $\mathbf{S}$ are independent

Results for ML Estimators and Large-Sample Properties II

Wishart distribution with m d.f. : Distribution of $\sum_{j=1}^m \mathbf{Z}_j \mathbf{Z}_j'$

where $\mathbf{Z}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$.

$\mathbf{X}_1, \dots, \mathbf{X}_n$ independent with mean $\boldsymbol{\mu}$ and finite variance - covariance $\mathbf{\Sigma}$
with Large n and $n - p$:

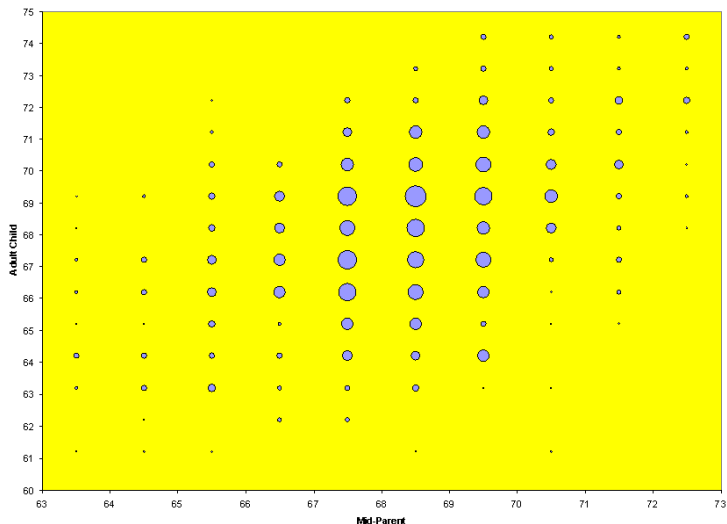
$\sqrt{n} (\bar{\mathbf{X}} - \boldsymbol{\mu})$ is approximately distributed $\mathcal{N}_p(\mathbf{0}, \mathbf{\Sigma})$

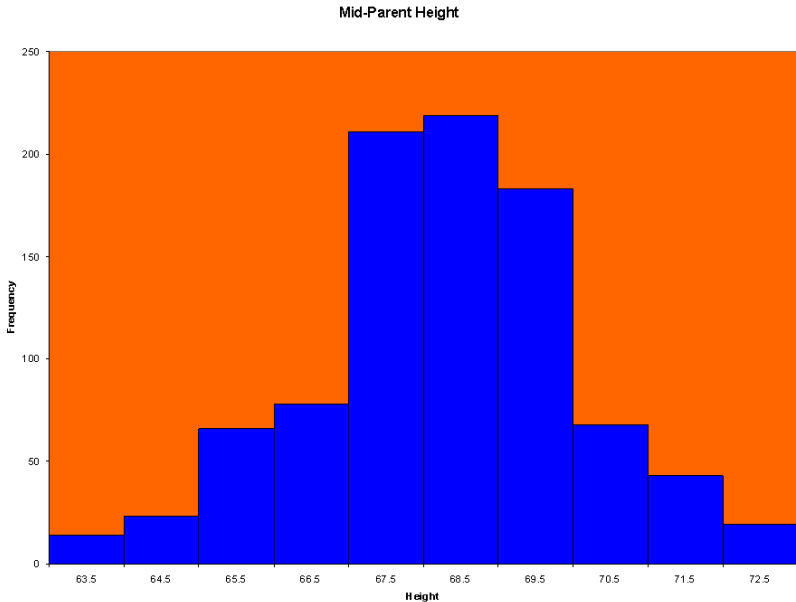
$n (\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu})$ is approximately distributed χ_p^2

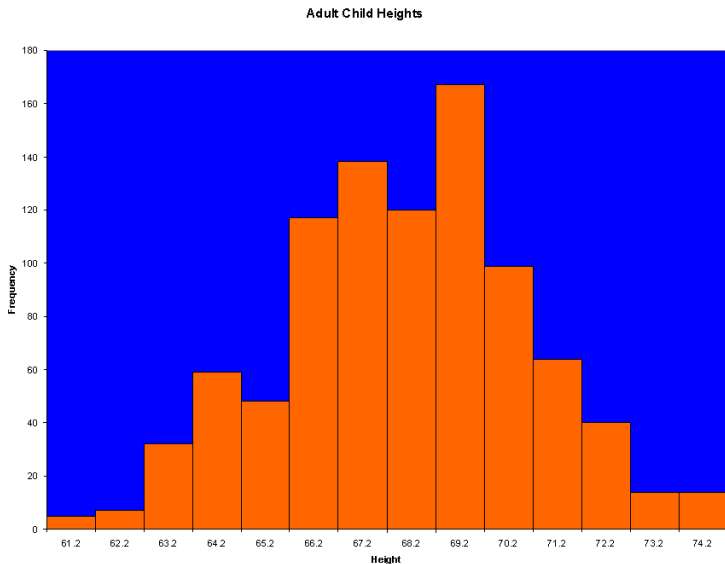
Data – Heights of Adult Children and Parents

- Adult Children Heights are reported by inch, in a manner so that the median of the grouped values is used for each (62.2", ..., 73.2" are reported by Galton).
 - He adjusts female heights by a multiple of 1.08
 - We use 61.2" for his "Below"
 - We use 74.2" for his "Above"
- Mid-Parents Heights are the average of the two parents' heights (after female adjusted). Grouped values at median (64.5", ..., 72.5" by Galton)
 - We use 63.5" for "Below"
 - We use 73.5" for "Above"

Adult Child vs Mid-Parent Height







Bivariate normal density

Joint Density Function: With $-\infty < x_1, x_2 < \infty$

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp \left\{ - \left(\frac{1}{2(1-\rho^2)} \right) \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right] \right\},$$

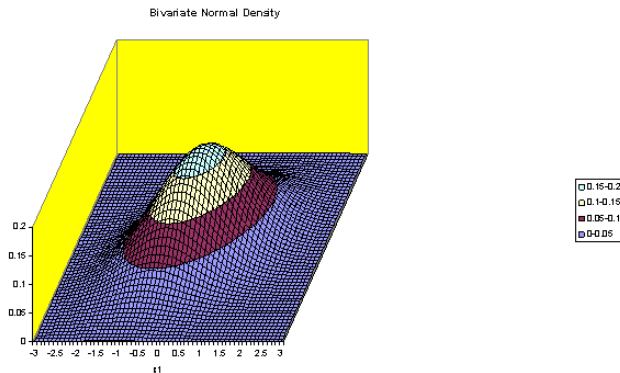
where :

$$\mu_1 = E\{X_1\} \quad \sigma_1^2 = V\{X_1\}$$

$$\mu_2 = E\{X_2\} \quad \sigma_2^2 = V\{X_2\}$$

$$\rho = \frac{E\{(X_1-\mu_1)(X_2-\mu_2)\}}{\sigma_1\sigma_2}$$

Bivariate normal density



Marginal Distribution of X

$$\begin{aligned} f_1(x_1) &= \int_{-\infty}^{\infty} f(x_1, x_2) dx_2 \\ &= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left\{-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right\} \int_{-\infty}^{\infty} \exp\left\{-\left[\frac{x_2 - \left[\mu_2 + (x_1 - \mu_1)\left(\frac{\sigma_2}{\sigma_1}\right)\rho\right]}{2(1-\rho^2)\sigma_2^2}\right]^2\right\} dx_2 \end{aligned}$$

The integrand is proportional to a normal density with:

$$E\{X_2\} = \mu_2 + (x_1 - \mu_1)\left(\frac{\sigma_2}{\sigma_1}\right)\rho \quad V\{X_2\} = (1 - \rho^2)\sigma_2^2$$

Taking the normalizing constant from the constant in front gives us:

$$\begin{aligned} f_1(x_1) &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right\} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi(1-\rho^2)\sigma_2^2}} \exp\left\{-\left[\frac{x_2 - \left[\mu_2 + (x_1 - \mu_1)\left(\frac{\sigma_2}{\sigma_1}\right)\rho\right]}{2(1-\rho^2)\sigma_2^2}\right]^2\right\} dx_2 \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right\}. \end{aligned}$$

Conditional Distribution of X_1 Given $X_2 = x_2$

$$\begin{aligned} f(x_1|x_2) &= \frac{f(x_1, x_2)}{f_1(x_2)} \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2(1-\rho^2)}} \exp \left\{ - \left(\frac{1}{2(1-\rho^2)\sigma_1^2} \right) \left[x_1 - \left(\mu_1 + \frac{(x_2 - \mu_2)\rho\sigma_1}{\sigma_2} \right) \right]^2 \right\} \\ \Rightarrow X_2|X_1 = x_1 &\sim \mathcal{N} \left[\mu_1 + \frac{(x_2 - \mu_2)\rho\sigma_1}{\sigma_2}, \sigma_1^2 (1 - \rho^2) \right] \end{aligned}$$

Conditional Distribution of X_2 Given $X_1 = x_1$

$$\begin{aligned} f(x_2|x_1) &= \frac{f(x_1, x_2)}{f_1(x_1)} = \\ &= \frac{1}{\sqrt{2\pi\sigma_2^2(1-\rho^2)}} \exp \left\{ - \left(\frac{1}{2(1-\rho^2)\sigma_2^2} \right) \left[x_2 - \left(\mu_2 + \frac{(x_1 - \mu_1)\rho\sigma_2}{\sigma_1} \right) \right]^2 \right\} \\ \Rightarrow X_2|X_1 = x_1 &\sim \mathcal{N} \left[\mu_2 + \frac{(x_1 - \mu_1)\rho\sigma_2}{\sigma_1}, \sigma_2^2 (1 - \rho^2) \right] \end{aligned}$$

This is referred to as the REGRESSION of X_2 on X_1

Summary of Results

Joint Distribution: $f(x_1, x_2) =$

$$\frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp \left\{ - \left(\frac{1}{2(1-\rho^2)} \right) \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right] \right\}$$
$$-\infty < x_1, x_2 < \infty$$

Marginal Distributions:

$$f_1(x_1) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp \left\{ - \frac{(x_1-\mu_1)^2}{2\sigma_1^2} \right\} \quad -\infty < x_1 < \infty \quad X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$$

$$f_2(x_2) = \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp \left\{ - \frac{(x_2-\mu_2)^2}{2\sigma_2^2} \right\} \quad -\infty < x_2 < \infty \quad X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$$

Conditional Distributions:

$$f(x_2|x_1) = \frac{1}{\sqrt{2\pi\sigma_2^2(1-\rho^2)}} \exp \left\{ - \left(\frac{1}{2(1-\rho^2)\sigma_2^2} \right) \left[x_2 - \left(\mu_2 + \frac{(x_1-\mu_1)\rho\sigma_2}{\sigma_1} \right) \right]^2 \right\}$$

$$f(x_1|x_2) = \frac{1}{\sqrt{2\pi\sigma_1^2(1-\rho^2)}} \exp \left\{ - \left(\frac{1}{2(1-\rho^2)\sigma_1^2} \right) \left[x_1 - \left(\mu_1 + \frac{(x_2-\mu_2)\rho\sigma_1}{\sigma_2} \right) \right]^2 \right\}$$

$$X_2|X_1 = x_1 \sim \mathcal{N} \left[\mu_2 + \frac{(x_1-\mu_1)\rho\sigma_2}{\sigma_1}, \sigma_2^2(1-\rho^2) \right]$$

$$X_1|X_2 = x_2 \sim \mathcal{N} \left[\mu_1 + \frac{(x_2-\mu_2)\rho\sigma_1}{\sigma_2}, \sigma_1^2(1-\rho^2) \right]$$

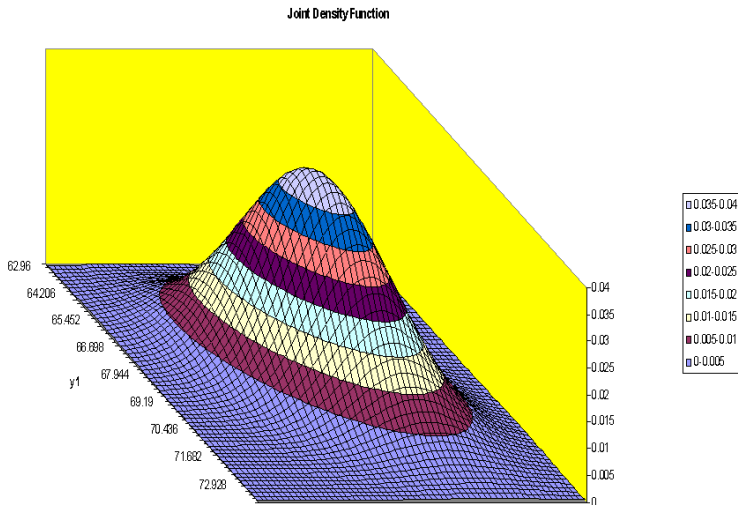
Heights of Adult Children and Parents

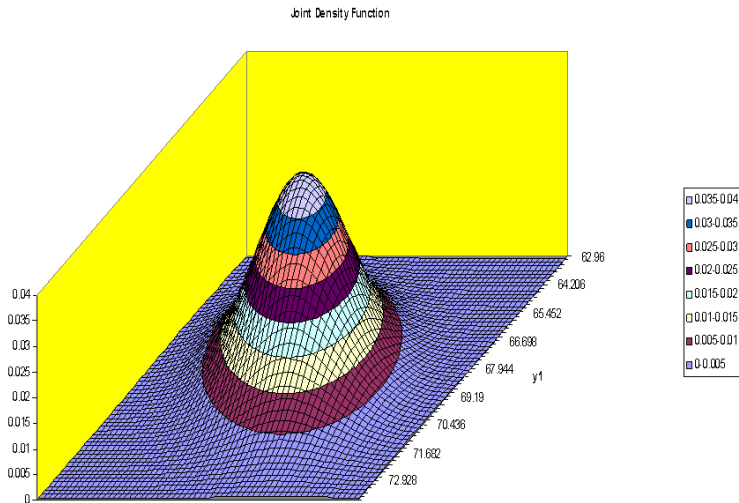
- Empirical Data Based on 924 pairs (F. Galton)
- X_2 = Adult Child's Height
 - $X_2 \sim \mathcal{N}(68.1, 6.39)\sigma_2 = 2.53$
- X_1 = Mid-Parent's Height
 - $X_1 \sim \mathcal{N}(68.3, 3.18)\sigma_1 = 1.78$
- $\text{COV}(X_1, X_2) = 2.02 \Rightarrow \rho = 0.45, \rho^2 = 0.20$
- $X_2|X_1 = x_1$ is Normal with conditional mean and variance:

$$E[X_2|X_1 = x_1] = \mu_2 + \frac{(x_1 - \mu_1)\rho\sigma_2}{\sigma_1} = 68.1 + \frac{(x_1 - 68.3)(0.45)\sqrt{6.39}}{\sqrt{3.18}} = 68.1 + 0.638x_1 - 43.6 = 24.5 + 0.638x_1$$

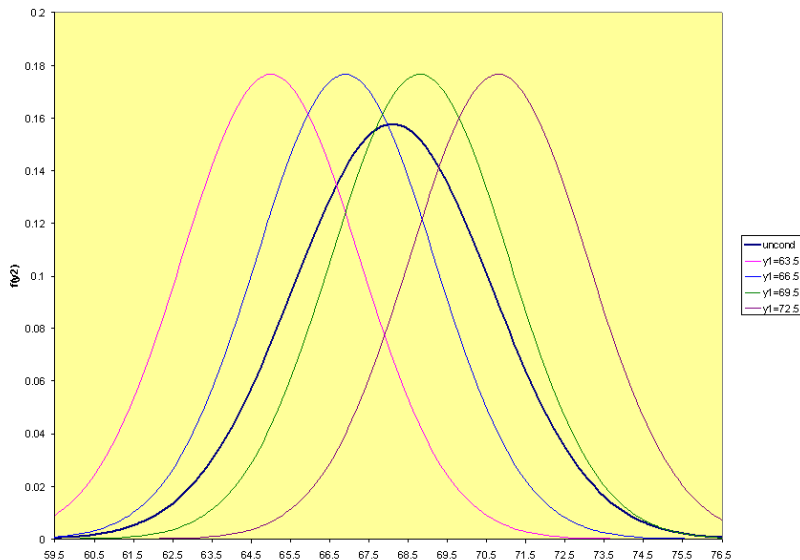
$$V[X_2|X_1 = x_1] = \sigma_2^2(1 - \rho^2) = 6.39(1 - 0.20) = 5.11 \quad \sigma_{Y_2|Y_1} = \sqrt{5.11} = 2.26$$

x1	Unconditional	63.5	66.5	69.5	72.5
E[X ₂ x ₁]	68.1	65.0	66.9	68.8	70.8
sX ₂ x ₁	2.53	2.26	2.26	2.26	2.26

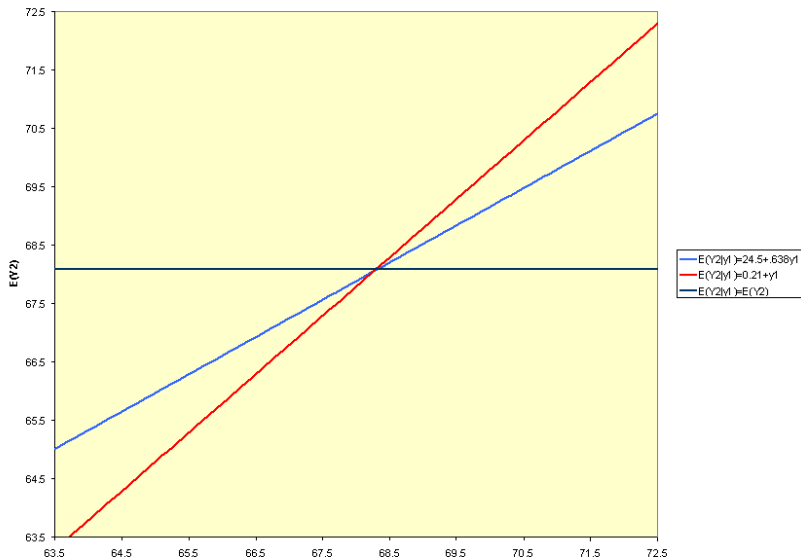




Distributions of Heights of Adult Children



Regression to the Mean



Expectations and Variances

- $E\{X_1\} = 68.3$ $V\{X_1\} = 3.18$
- $EX_2 = 68.1$ $VX_2 = 6.39$
- $EX_2|X_1 = x_1 = 24.5 + 0.638x_1$
- $E_{X_1}[E\{X_2|X_1 = x_1\}] = E_{X_1}[24.5 + 0.638X_1] = 24.5 + 0.638(68.3) = 68.1 = E\{X_2\}$
- $V\{X_2|X_1 = x_1\} = 5.11 \Rightarrow E_{X_1}[V\{X_2|X_1 = x_1\}] = 5.11$
- $V_{X_1}[E\{X_2|X_1 = x_1\}] = V_{X_1}[24.5 + 0.638X_1] = (0.638)^2 V(X_1) = (0.407)3.18 = 1.29$
- $E_{X_1}[VX_2|X_1 = x_1] + V_{Y_1}[E\{X_2|X_1 = x_1\}] = 5.11 + 1.29 = 6.40 = V(X_2)$ (with round-off)